How efficiently can we describe mathematical phenomena?

Bruno de Mendonça Braga

IMPA

May 26th, 2025

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- Description of dualization in Banach spaces in a computable way.

  (Ann. Inst. Fourier.)

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(Suslin, 1917)

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A sequence $(x_i)_i$ is a **basis** for a X if for all $x \in X$ there is a unique $(\alpha_i)_i \in \mathbb{R}^{\mathbb{N}}$ such that

$$x = \sum_{i=1}^{\infty} \alpha_i x_i.$$

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Conjecture I

The class of separable Banach spaces admitting a basis is analytic non-Borel.

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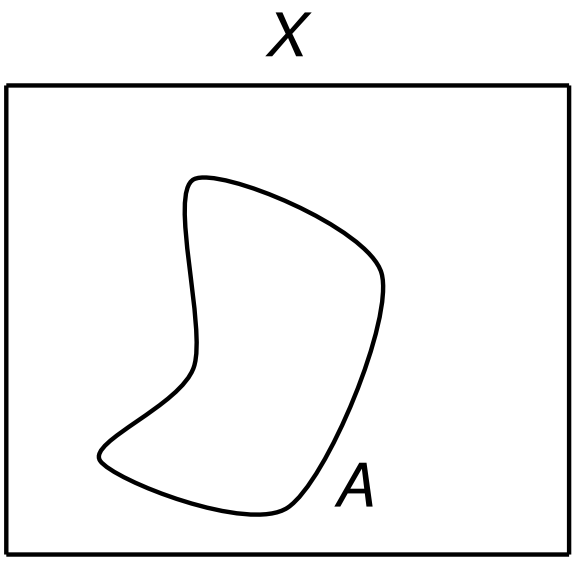

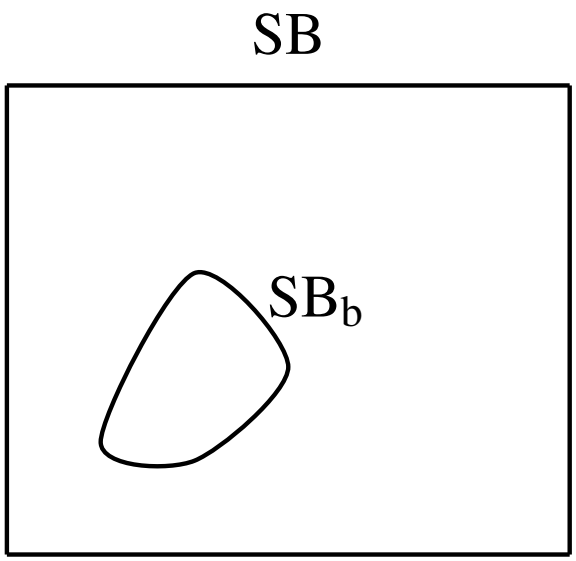

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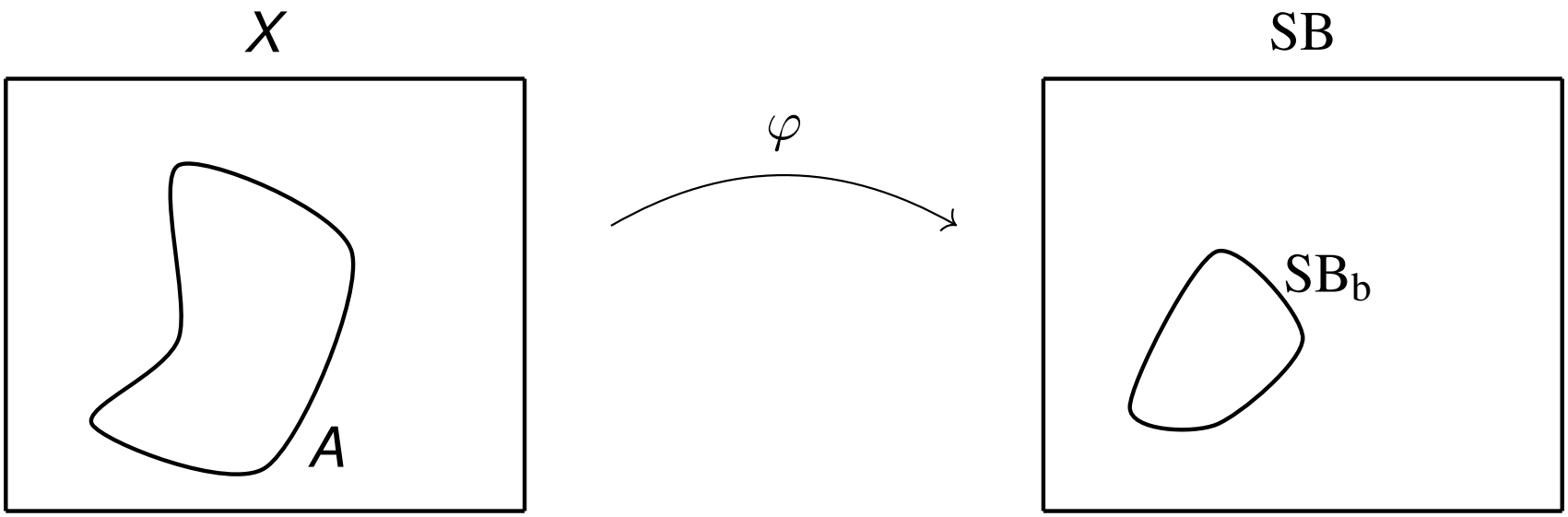

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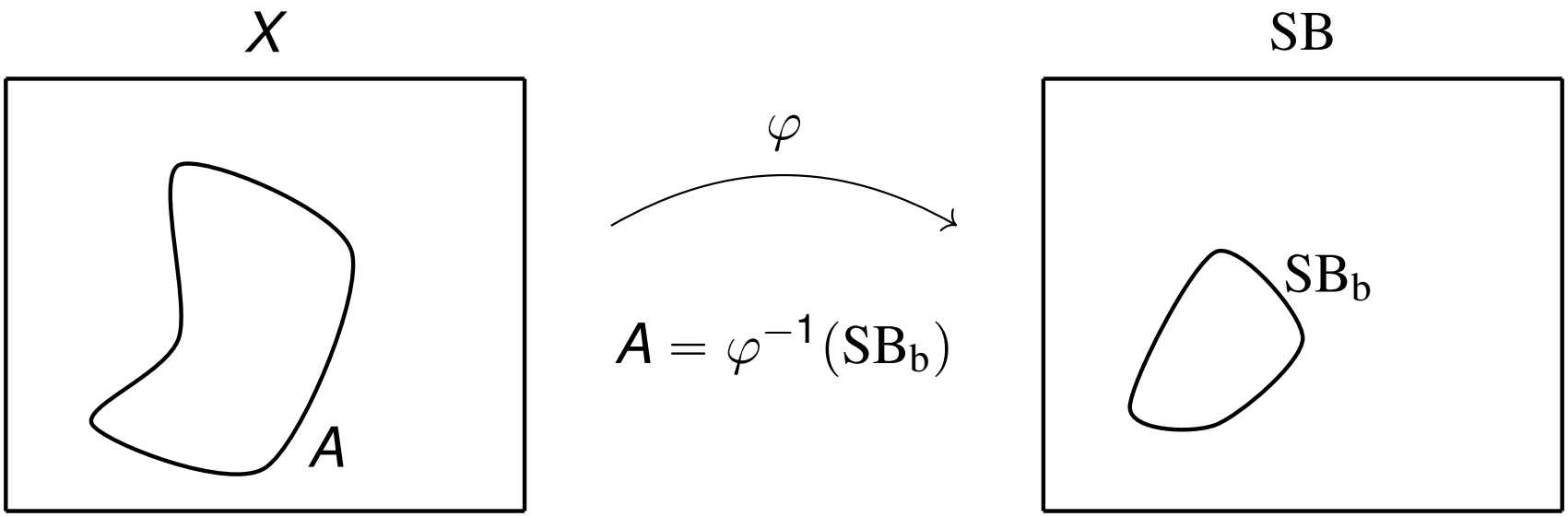

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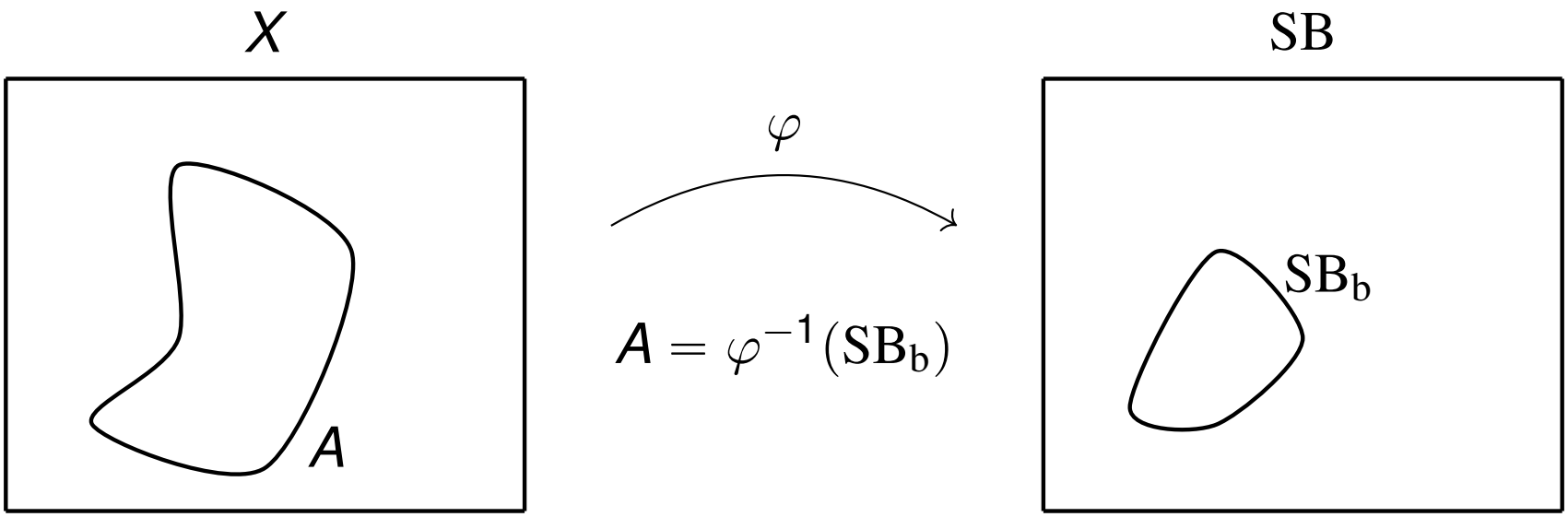

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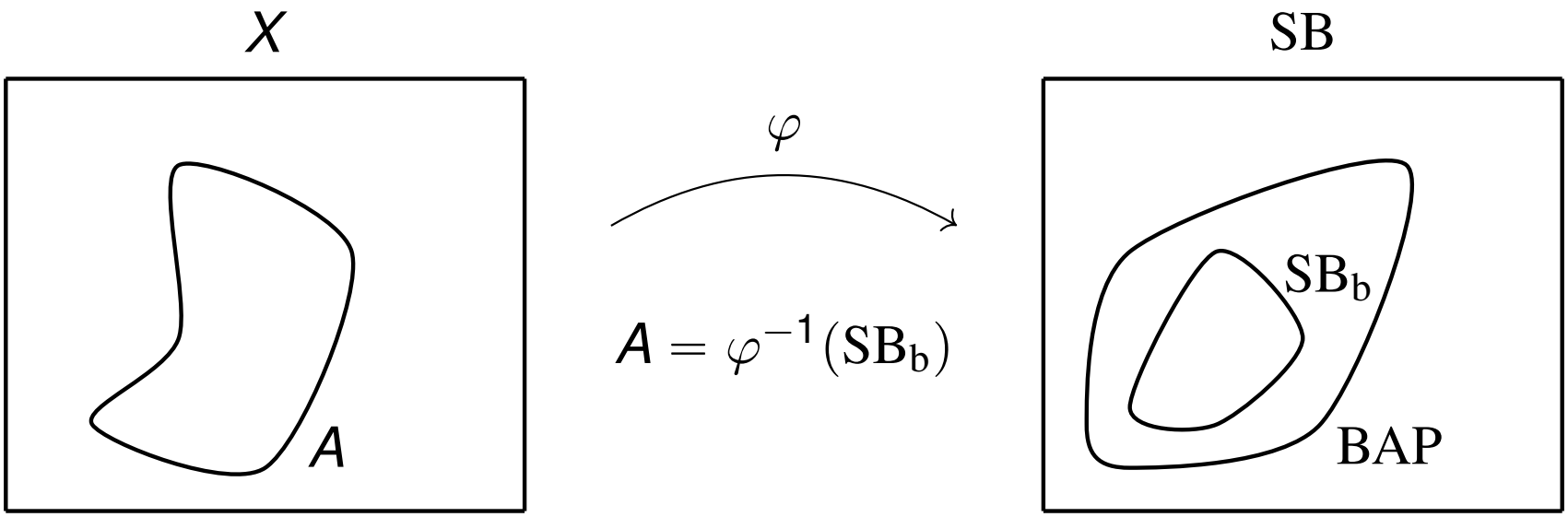

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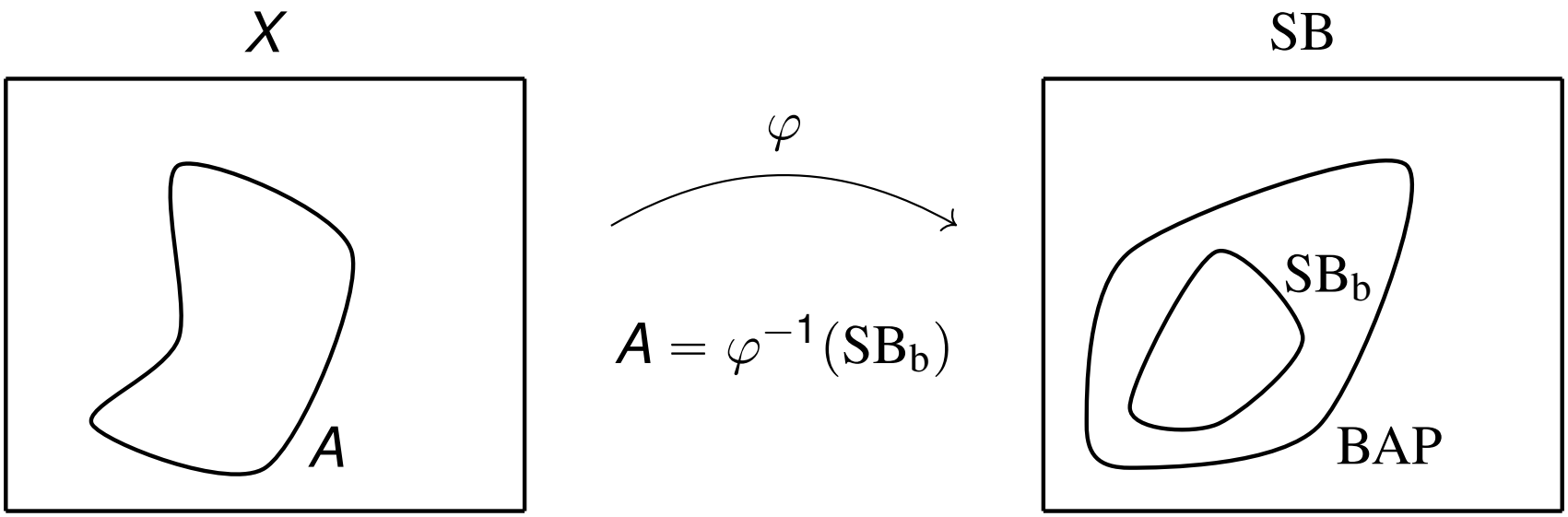

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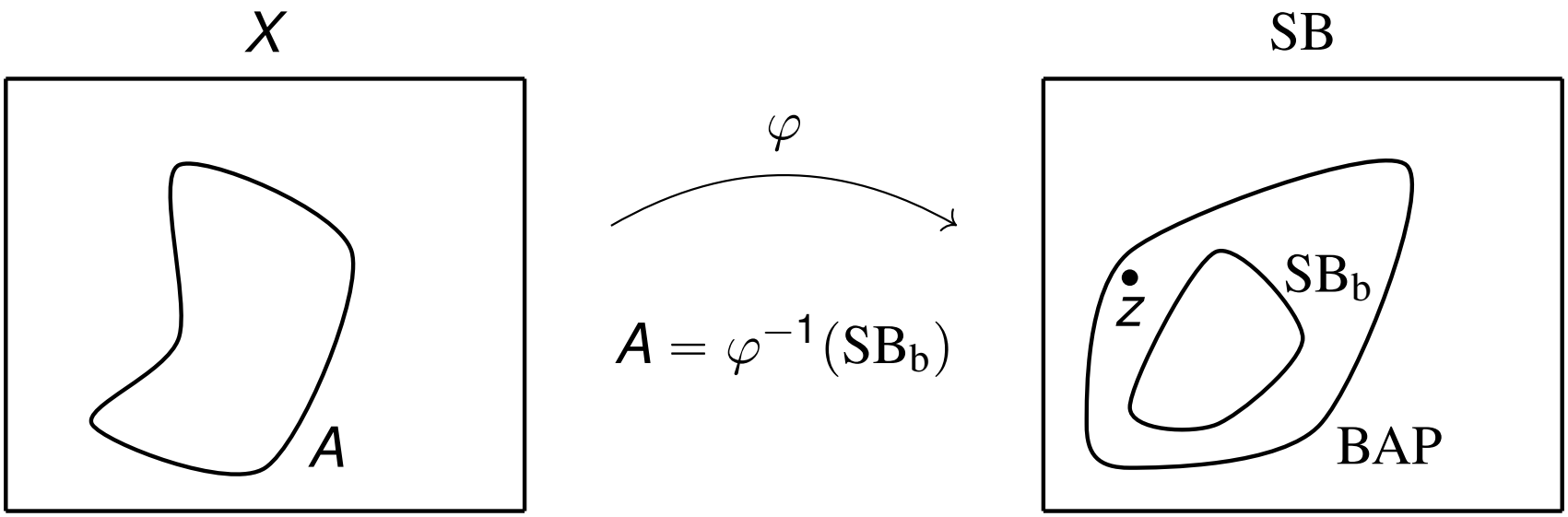

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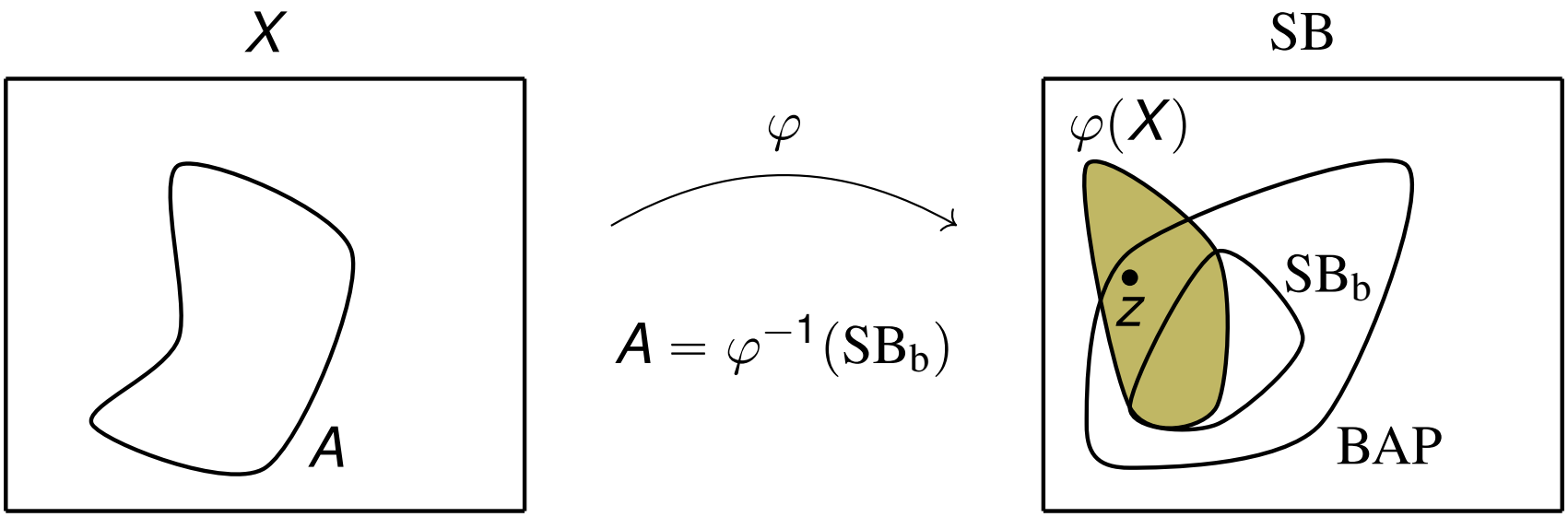

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Conjecture III

The class of all separable Montel spaces has no universal element.

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- Bringing complexity to Fréchet spaces has potencial of revolutionizing the field

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(Muito obrigado!)